Naveenkumar Narsozhan

Los Angeles, CA — 213-796-2976 — narsozha@usc.edu — Portfolio — LinkedIn — GitHub

Education

University of Southern California, Los Angeles, CA Aug 2025 – Expected May 2027

Master of Science in Computer Science – ***CGPA: 3.82/4.0***

- Relevant Coursework: Analysis of Algorithms, Information Retrieval, Natural Language Processing, Database Systems

National Institute of Technology Puducherry, India Dec 2020 – May 2024

B.Tech in Computer Science and Engineering – ***CGPA: 9.3/10.0***

Technical Skills

Languages: Python, C++, C#, SQL, JavaScript, TypeScript
ML/AI: PyTorch, TensorFlow, scikit-learn, Hugging Face Transformers, OpenCV, YOLO, LangChain
Backend & Data: NumPy, pandas, FastAPI, ASP.NET Core, Node.js, PostgreSQL, MongoDB, Chroma, Supabase
Cloud & Tools: Azure, Docker, Git, Linux, Vercel, Azure DevOps

Professional Experience

Microchip Technology Inc. May 2026 – Aug 2026

Application Engineering Intern — SSD Software *Roseville, CA*

- Built a cross-platform Python GUI using CustomTkinter to configure and monitor prototype SSD firmware via PCIe/NVMe interfaces, enabling live technical demonstrations of two storage technologies to 500+ attendees at Future of Memory and Storage (FMS) 2026.
- Automated benchmark execution, telemetry collection, and failure-log analysis across 1,000+ PCIe/NVMe test runs, reducing manual testing time by 30% and accelerating the isolation of critical firmware defects.

HCLTech Sep 2024 – Jul 2025

Technical Lead - Azure Application Development *Madurai, India*

- Developed authentication and authorization features for an enterprise web portal using C#, ASP.NET Core, Entity Framework Core, JWT, and role-based access control, separating client and administrator workflows.
- Contributed to the migration of a legacy application serving 50K+ monthly users to Azure App Service and Azure SQL Database, configuring Azure DevOps CI/CD workflows for automated deployments.

National Institute of Ocean Technology May 2023 – Jun 2023

ML Research Intern *Chennai, India*

- Engineered features from 1M+ raw oceanographic data points using pandas and NumPy to map vessel trajectories and quantify the impact of environmental parameters on marine navigation.
- Trained and evaluated predictive machine learning models to determine optimal, fuel-efficient maritime routes by forecasting the effects of adverse sea conditions on vessel speed.

Projects

Dynamic Vision Inference Pipeline | *PyTorch, YOLOv11, Object Detection* Jan 2026 – Apr 2026

- Designed a confidence-based cascade between NanoDet-Plus and YOLO11L, using a validation-set threshold sweep to route low-confidence images to the higher-capacity detector.
- Evaluated on 36,800 X-ray images with COCO-format annotations; reduced inference time by 54% versus YOLOv11L-only execution while improving mAP@0.5 by 6.97 percentage points over NanoDet-Plus alone.

Kuri AI – LLM-Driven Task Understanding System | *Multimodal LLMs, Agentic AI* Nov 2025 – Mar 2026

- Engineered a voice-enabled task management application using Next.js, TypeScript, and the Web Speech API, implementing real-time client-side speech transcription and animated UI state transitions with Framer Motion and Tailwind CSS.
- Architected an API pipeline leveraging Google Gemini for structured zero-shot task categorization, designing a heuristic fallback engine and source-tracing system to ensure continuous operation during API rate limits or network outages.

VibeArchitect (Agentic Content Pipeline) | *Special Mention - Solo developer, TrojanHacks USC* Feb 2026

- Engineered a dual-agent workflow combining Gemini for prompt augmentation and FLUX.1 via the Hugging Face Inference API, integrating the Presaige API to drive an iterative feedback loop that autonomously refines thumbnails until predicted engagement thresholds are met.
- Implemented job-state polling, automated retry handling, and failure recovery for asynchronous scoring requests, completing all evaluated workflows without timeout failures.

Valorant Strategy Assistant - RAG Chatbot | *Semantic Search, ChromaDB, NLP* Sep 2025 – Dec 2025

- Developed a retrieval-augmented generation (RAG) pipeline integrating a quantized Mistral 7B LLM to provide context-aware Q&A capabilities based on localized game data.